

The Policy Triangle: A Taxonomy of Policy Mismatch in LLM Reinforcement Learning

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Project page: <https://huaiyizhao.github.io/policy-triangle/>

Abstract

Large-language-model reinforcement learning (LLM-RL) increasingly separates rollout generation from optimization, creating mismatch among the behavior policy that generated a token, the proximal policy that anchors an update, and the current policy being trained. Existing methods use importance weights, clipping, and rejection, but are difficult to compare because similar ratios play different roles. We organize these mechanisms with the *policy triangle*: for behavior μ , proximal q , and current π_θ , its edges are $B = q/\mu$, $U = \pi_\theta/q$, and $E = \pi_\theta/\mu = BU$. An intervention is described by an edge, an operator—detached mask, detached weight, or differentiable update shaper—and a token-, sequence-, or group-level geometry. A simple accounting check requires an unshaped behavior-to-current estimator to contain exactly one copy of E , which helps expose accidental double counting. Rather than proposing a new optimizer, the triangle provides a common taxonomy for direct and decoupled methods, clarifies how published approaches differ, and reveals combinations that may merit empirical study. The framework is conceptual and does not by itself establish new convergence or performance results.

1 Introduction

Reinforcement learning is central to language-model alignment and reasoning [3, 14, 32]. Classical policy gradients begin with REINFORCE [23]; TRPO and PPO add proximal control to permit stable reuse of sampled data [18, 19]. PPO and its group-relative variants remain widely used in LLM training [20, 25], while simpler REINFORCE-style objectives have regained attention [1, 8, 11, 13]. At the same time, modern systems separate inference from training to improve throughput [4, 21]. Rollouts may be stale, generated by a different numerical backend, or produced under low precision or mixture-of-experts routing. The training policy that scores those tokens can therefore disagree with the actual behavior policy even when nominal weights match [10, 12, 17, 31].

Recent methods address this mismatch through a rapidly expanding vocabulary. Decoupled PPO separates

behavior correction from proximal control [6]; A-3PO approximates the proximal anchor [9]; GSPO moves clipping to a sequence ratio [30]; rollout-correction systems combine truncated importance sampling (TIS) and rejection sampling [22]; IcePop and KPop mask discrepant tokens [5, 12]; DPPO replaces sampled-ratio clipping with divergence geometry [16]; and TRM masks sequences using a worst-token trust-region statistic [10]. These methods are typically presented as unrelated losses, obscuring whether an operation selects data, changes measure, or constrains a trainable update.

Our aim is to supply a common coordinate system rather than another optimizer. In the *policy triangle*, policies rather than named objectives are the anchors, and the three edges represent rollout mismatch, current update drift, and their total. Within this view, the paper:

- separates where a mismatch is measured from whether it is masked, reweighted, or used to shape a trainable update;
- states a bookkeeping test for missing or duplicated behavior-to-current ratios; and
- classifies representative methods and identifies several untested combinations suggested by the resulting design space.

The result is an analytical taxonomy. It offers notation, comparisons, and design hypotheses, but no new optimization guarantee or empirical superiority claim.

Relation to nearby unified views. The closest precursor is decoupled PPO, which already inserts a proximal policy to separate behavior correction from update control [6]; AReaL and A-3PO adapt that factorization to asynchronous LLM training [4, 9]. Our scope is broader in operator placement but shallower algorithmically: we classify masks, weights, shapers, and their geometries without proposing a particular proximal anchor. The veRL rollout-correction formulation and FP8-RL compare bypass/decoupled TIS, MIS, and rejection configurations [17, 22]; the triangle abstracts that implementation pipeline and additionally locates direct PPO-, GSPO-, and divergence-based shapers. RPG and the off-policy interpretation of group-relative REINFORCE provide first-principles analyses of importance weighting, KL

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regularization, baselines, and data shaping [24, 27]; our view is narrower—it does not unify advantage or reward-regularization paths—but makes the three policy identities and operator geometry explicit. Broader post-training taxonomies organize on/off-policy learning by trajectory provenance and behavioral role [29]; the triangle instead resolves the policy-loss correction stage at finer granularity. Finally, TRM, ESTR, and Jackpot are concrete algorithms for particular trust-region or rejection mechanisms [2, 10, 28]; here they are comparison points, not consequences claimed by the taxonomy.

2 The Policy Triangle

For prompt x , response $y = (y_1, \dots, y_T)$, and context $h_t = (x, y_{<t})$, define

$$\mu_t = \mu(y_t | h_t), \quad q_t = q(y_t | h_t), \quad \pi_{\theta_t} = \pi_{\theta}(y_t | h_t). \quad (1)$$

The three token ratios are

$$B_t = \frac{q_t}{\mu_t}, \quad U_t = \frac{\pi_{\theta_t}}{q_t}, \quad E_t = \frac{\pi_{\theta_t}}{\mu_t} = B_t U_t. \quad (2)$$

The AB edge in Figure 1 records mismatch already present when data reaches the learner: staleness and backend disagreement. The BC edge records update drift introduced by optimization. The direct AC edge records their total.

Bypass and factorization. In bypass mode, $q \equiv \mu$, hence $B = 1$ and $U = E$. Statistical correction and proximal control are forced onto the same edge. In decoupled mode, q is independently defined and frozen during an update; B can correct rollout mismatch while U constrains the current step [4, 6]. The proximal anchor is itself a design variable: A-3PO uses a staleness-aware interpolation between behavior and current log-probabilities while preserving $E = BU$ [9].

2.1 Three edge operators

For any edge $X \in \{B, U, E\}$, we distinguish operators by gradient semantics.

Mask. $M_{g,s}(X, \hat{A}) \in \{0, 1\}$ is detached. It removes a token, sequence, or group according to geometry g and statistic s . Ratio bands, K2/K3 gates, binary-KL masks, and full-distribution trust regions are examples. A mask consumes no density ratio, but introduces selection bias.

Weight.

$$W_{g,s}(X) = \text{sg}[f_{g,s}(X)] \geq 0 \quad (3)$$

is a detached multiplier. Raw, truncated, tapered, and self-normalized IS are examples. A ratio-carrying weight consumes its edge in the change of measure.

Differentiable shaper. A shaper enters the loss through π_{θ} . PPO’s effective coefficient on edge X is

$$C^{\text{PPO}}(X, \hat{A}) = X \mathbf{1} \left[\begin{array}{l} \hat{A} \geq 0, \quad X \leq 1 + \epsilon_+, \text{ or } \\ \hat{A} < 0, \quad X \geq 1 - \epsilon_- \end{array} \right]. \quad (4)$$

Unlike truncating a detached weight, selecting the clipped branch removes gradient in the blocked direction. DAPO changes the sign-dependent bounds [25]; GSPO replaces the token ratio with a length-normalized sequence ratio [30].

An algorithm need not have one global granularity. It may use token TIS, a sequence K3 mask, and token PPO clipping simultaneously. Each attachment independently selects a statistic: token ratio; sequence product $R_X = \prod_t X_t$; geometric ratio $G_X = \exp(T^{-1} \sum_t \log X_t)$; group aggregation; or a sampled/full-distribution divergence. Sequence-product IS is an exact trajectory change of measure but high variance; G_X is length-normalized but is not a trajectory density ratio.

These choices admit a common gradient-kernel notation. Let i index the unit selected by an operator’s geometry. A broad class of policy-loss gradients is

$$\hat{g} = \frac{1}{Z} \sum_i M_i \bar{W}_i C_i \hat{A}_i \nabla_{\theta} \log \pi_{\theta}(y_i | h_i), \quad (5)$$

where an absent operator is the multiplicative identity. Equation (5) describes effective gradient coefficients rather than requiring every method to share one scalar loss. A concrete attachment is fully specified by (edge, operator, scope, statistic, sign rule, normalization).

3 Ratio Accounting

The triangle supplies a simple correctness audit. Consider the *unshaped limit*: masks equal one, truncation and normalization are disabled, and clipping lies on its linear branch.

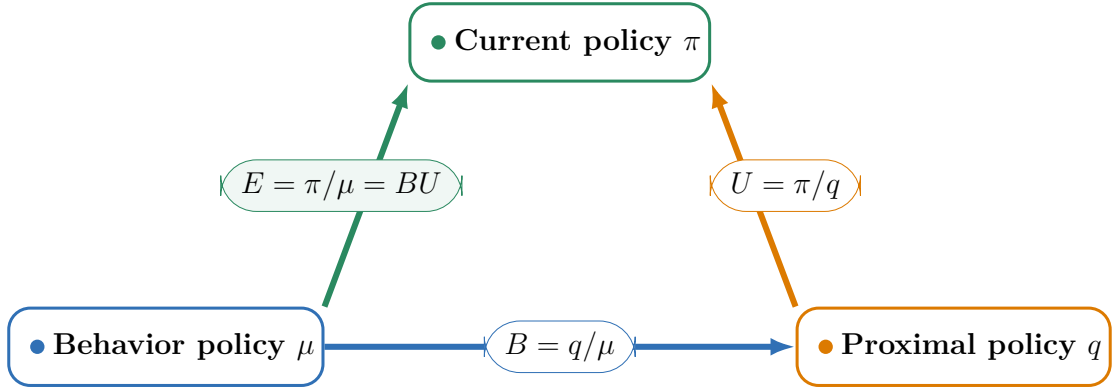
Accounting check. Call an estimator *ratio complete* when its unshaped coefficient equals the direct behavior-to-current ratio exactly once. A direct construction then consumes E once, while a factorized construction consumes B and U once each:

$$K^0 = E = BU. \quad (6)$$

This is bookkeeping rather than a new theorem: the sampled-action change of measure is $\pi_{\theta}/\mu = E$, and inserting q gives $(q/\mu)(\pi_{\theta}/q) = BU$. An extra factor changes the measure; an omitted factor leaves an edge uncorrected.

Consequently, $C(E)$, $W(E)$, $W(B)C(U)$, and $W(B)W(U)$ are ratio complete in their linear limits. In contrast,

$$W(E)C(E) \rightsquigarrow E^2, \text{quad} W(E)C(U) \rightsquigarrow BU^2, \quad (7)$$



Every edge may host one of three operators; each attachment independently chooses token, sequence, or group geometry.



Figure 1: **The policy triangle.** Behavior μ , proximal q , and current π_θ are policy anchors; $E = BU$. Each edge may host a detached mask, a detached weight, or a differentiable clipping/shaping operator. Geometry is selected per attachment. Bypass identifies $q \equiv \mu$, so $B = 1$ and $U = E$.

which double-count ratios. This explains why bypass PPO must not multiply an additional $W(E)$: its clipped loss already carries E .

Ratio completeness is deliberately narrower than unbiasedness. A ratio-complete estimator can still be biased by clipping, masking, token-level approximation, or self-normalization, and it may have unusable variance. The check only detects a structural class of implementation errors: consuming an edge twice or omitting it before any intended bias–variance transformation is applied.

Nonlinear placement matters. Although $E = BU$, nonlinear operators do not distribute:

$$\text{clip}(E) \neq \text{clip}(B) \text{clip}(U), \quad \mathbf{M}(E) \neq \mathbf{M}(B)\mathbf{M}(U). \quad (8)$$

For $k_3(x) = x - 1 - \log x$,

$$k_3(BU) = k_3(B) + k_3(U) + (B - 1)(U - 1). \quad (9)$$

Thus an end-to-end K3 filter is not equivalent to independent filters on rollout mismatch and update drift.

Variance control. TIS replaces w_i by $\tilde{w}_i = \min(w_i, c)$ [7]. Self-normalization is orthogonal:

$$\bar{w}_i = \frac{\tilde{w}_i}{\sum_j \mathbf{M}_j \tilde{w}_j / \sum_j \mathbf{M}_j}. \quad (10)$$

It stabilizes mean gradient scale but is biased at finite sample size [15]. Its domain—token, sequence, prompt group, or batch—is part of the algorithm. Sign-conditioned thresholds, as in DAPO and TOPR, are likewise operator parameters rather than new topology [8, 25].

4 Existing Methods on the Triangle

Table 1 maps representative policy-loss mechanisms. The table isolates policy mismatch; it does not enumerate reward shaping, dynamic sampling, entropy bonuses, or systems features.

The triangle clarifies three recurring ambiguities. First, bypass is not “IS disabled”: $B = 1$ and the direct loss still contains $E = U$. Second, PPO and GSPO are not pre-filters. Their clipped branches can yield zero gradient in an advantage-dependent direction, whereas a mask rejects independently of the update direction. Third, a decoupled filter may read any raw edge. $\mathbf{M}(B)$ is a static admission test when q is frozen; $\mathbf{M}(U)$ or $\mathbf{M}(E)$ is generally dynamic because π_θ changes. Recent diagnostics confirm that back-end mismatch and optimization drift can interact rather than behave as one static error source [26, 28, 31].

Methods omitted from the table also obtain simple signatures. ReMax and RLOO are direct policy-gradient baselines with no PPO shaper [1, 11]; adding off-policy data requires an explicit $W(E)$ or a factorized alternative. Conversely, using a separate $W(E)$ with an ordinary direct PPO loss is not a more complete correction—it is the E^2 error identified in Section 3. This distinction is especially important in configurable systems where rollout correction and policy loss are selected independently.

Jackpot illustrates a boundary of this loss-centered representation: its rejection sampler and jointly updated rollout model actively change the behavior distribution before a trajectory reaches the learner [2]. The triangle can label the resulting behavior policy and downstream correction,

Table 1: Representative placements. M, W, and C denote mask, detached weight, and differentiable shaper. Geometry is shown as a subscript where central to the method.

Method family	Topology	Triangle placement	Interpretation
PPO / GRPO / DAPO [19, 20, 25]	Coupled	$C_{\text{tok}}(E)$	One direct ratio carries both correction and proximal-control roles; DAPO uses asymmetric bounds.
CISPO / TOPR [8, 13]	Direct PG	$W_{\text{tok}}(E)$	Detached clipped or tapered weight; gradients need not vanish outside a PPO band.
GSPO [30]	Coupled	$C_{\text{geo}}(E)$	Sequence-geometric ratio performs sequence-level clipping; no separate filter in the original objective.
Decoupled PPO / AReaL / A-3PO [4, 6, 9]	Factorized	$W(B)C(U)$	Detached behavior correction is separated from the trainable proximal constraint.
TIS / MIS / rollout correction [7, 17, 22]	Either	$M(B \text{ or } E)W(B \text{ or } E)$, optional C	Token/sequence weighting and rejection can be combined; direct and decoupled modes change the edge semantics.
IcePop / KPop [5, 12]	Usually direct	$M_{\text{tok}}(E)$ plus a base loss	IcePop uses a ratio region; KPop uses bidirectional binary-KL geometry sensitive to absolute token probability.
DPPO / TRM [10, 16]	Direct	divergence $M(E)$ or divergence-based $C(E)$	Replace sampled-ratio magnitude with binary/top- k /full-distribution geometry; TRM rejects an entire sequence on worst-token divergence.

but it does not describe the complete sampling-control loop. Likewise, RPG’s treatment of KL-regularized objectives and group-relative REINFORCE’s analysis of baselines and data shaping cover axes intentionally outside the present policy-mismatch taxonomy [24, 27].

5 Design Implications and Scope

The triangle suggests a useful design heuristic: place detached correction near the behavior anchor and differentiable control near the current anchor. Exogenous mismatch on B has already occurred and can only be selected or reweighted; U is produced by the current optimizer and is therefore a natural trust-region edge. Bypass sacrifices this attribution but saves the proximal forward pass and can be effective in low-staleness regimes.

Two extensions preserve the heuristic. First, partial rollouts may use a token-indexed behavior version $v(t)$:

$$\mu(y | x) = \prod_t \mu_{v(t)}(y_t | h_t), \quad B_t = \frac{q_t}{\mu_{v(t),t}}. \quad (11)$$

Stored per-token sampling log-probabilities are sufficient for the observed conditional likelihood ratio, but not for unlogged prompt, replay, or resume-selection bias. Second, a general mask may read multiple edges and metadata,

$$M_i = \mathcal{F}(B_i, U_i, E_i, \text{age}_i, \text{version}_i, \text{sign } \hat{A}_i), \quad (12)$$

without forcing downstream IS to use the same edge.

Taxonomy-derived research hypotheses. To the best of our literature search through July 2026, the following exact combinations have not been evaluated as named

methods in the cited work. They are design templates, not claims of priority or effectiveness:

1. *Decoupled sequence shaping*: token-level rollout correction followed by a sequence-geometric proximal shaper, $W_{\text{tok}}(B)C_{\text{geo}}(U)$. This combines the attribution of decoupled PPO with the sequence coherence of GSPO without clipping the total ratio E .
2. *Heterogeneous two-edge control*: an entropy-adaptive admission mask on B , token correction $W(B)$, and a divergence-based mask plus ordinary shaper on U . The behavior edge can then tolerate exploratory high-entropy deviations differently from low-entropy back-end noise, while the update edge uses a distributional trust-region statistic.
3. *Segment-aware asynchronous correction*: for partial rollouts generated by several policy versions, normalize or truncate $W(B_t)$ within version-homogeneous segments, apply a sequence gate that may read total mismatch E , and retain $C(U)$ for the current update. This treats policy age as structured metadata rather than a single trajectory label.

A mask based only on frozen B can be computed once when a batch enters training, whereas a mask on U or E must generally be recomputed as π_θ changes.

For implementation, we recommend a five-step audit: (i) identify the actual token-level behavior distribution, including sampling transforms; (ii) identify the proximal anchor; (iii) record edge, detachment, statistic, geometry, sign dependence, and normalization for every operator; (iv) verify that the unshaped kernel contains exactly one

E ; and (v) compute filter statistics from raw ratios before truncation hides the tail. The audit distinguishes an intentional bias–variance trade-off from a structural E^2 error.

The taxonomy does not choose thresholds, guarantee that masking improves return, or repair support mismatch. It omits reward-model drift, environment nonstationarity, and optimizer-state staleness. Sequence-product IS can be formally exact and statistically unusable, while geometric statistics are stable but not exact change-of-measure factors. Empirical validation should instrument B , U , and E separately and report reward, effective sample size, rejection, gradient variance, policy age, and throughput.

6 Conclusion

The behavior, proximal, and current policies form a triangle whose edges satisfy $E = BU$. Masks select regions, detached weights change measure, and differentiable shapers control updates; each operator chooses its own geometry. Bypass collapses the triangle, while decoupling separates exogenous rollout mismatch from endogenous update drift. Requiring the unshaped kernel to recover exactly one copy of E provides a compact test for missing or double-counted correction. The main value of the triangle is comparative: it places many LLM-RL mechanisms in one coordinate system and makes underexplored, testable combinations easier to state without presenting them as established improvements.

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